# Navneet Kumar Mishra

ECE Undergraduate | Robotics, Embedded Systems & IoT

info.navneet.net@gmail.com | +91 7482037075
Patsa, Hasanpur, Samastipur, Bihar (848206)
LinkedIn | GitHub | Portfolio



## Professional Summary

ECE undergraduate (B.Tech, 2028) specialising in IoT-enabled robotics and embedded systems. Builds systems end to end: **SOLIDWORKS** design, **3D printing**, **ESP32** firmware in Embedded C, wireless dashboards and live video. Student Coordinator at e-Yantra Robotics Lab and former BSNL intern, seeking robotics and embedded internships.

## Education

**B.Tech, Electronics & Communication Engineering** — **2024 – 2028 (Expected)**
Government Engineering College Kishanganj | Currently in 5th Semester | **CGPA: 7.56** (up to 3rd Sem)

**Intermediate (XII), BSEB** — **2024**
Samastipur College, Samastipur | 67.4%

**Matriculation (X), CBSE** — **2022**
Progressive Central High School, Patsa | 86.4%

## Technical Skills

| | |
|---|---|
| **Programming Languages** | C, C++, Python |
| **Embedded Systems** | Embedded C, Arduino, ESP32, STM32, Raspberry Pi |
| **Communication Protocols** | UART, $I^2C$, SPI, MQTT |
| **Robotics & Automation** | ROS2, IoT, Sensor & Actuator Integration, Autonomous Robotics |
| **CAD, Simulation & Design** | SOLIDWORKS, MATLAB, 3D Printing, Proteus Design |
| **Development Tools** | Git, GitHub, Arduino IDE, STM32CubeIDE, PlatformIO, VS Code |
| **Areas of Interest** | Robotics, Embedded Systems, Industrial Automation, Drone Systems |

## Experience & Leadership

**Student Coordinator** — **Feb 2026 – Present**
e-Yantra Robotics Lab | Government Engineering College Kishanganj

- Coordinate day-to-day robotics lab activities and organise technical sessions for interested students.
- Mentor peers in Embedded Systems, Arduino and Robotics through hands-on experiments and workshops.
- Support student participation in robotics competitions and technical events.

**Technical Intern** — **Dec 2025 – Jan 2026**
Bharat Sanchar Nigam Limited (BSNL)

- Gained hands-on exposure to telecommunication networks, optical fibre communication, switching and transmission systems.

## Projects

**IoT Controlled Mobile Robotic Arm** — **Feb 2026 – May 2026**
*ESP32 | SOLIDWORKS | Embedded C | IoT | ESP32-CAM*

- Designed a custom robotic arm in **SOLIDWORKS** and built it with **3D printing** at the e-Yantra Robotics Lab.
- Engineered an ESP32-powered mobile platform integrating sensors and actuators for remote operation.
- Built a responsive web dashboard for wireless control and **real-time system monitoring**.
- Integrated ESP32-CAM for live **video streaming** and remote visual feedback.
- **Applications:** remote material handling, industrial automation, hazardous environments.

**IoT Based Metal Detection Robot** — **Feb 2026 – May 2026**
*ESP32 | Embedded C | Sensors | IoT*

- Developed an ESP32-powered robot that detects metal objects and reports them wirelessly.
- Integrated a metal-detection sensor, mobile platform and web interface for **real-time** operation and remote control.
- Implemented instant detection alerts and remote monitoring on a wireless dashboard.
- Optimised the embedded firmware for reliable detection, navigation and **low-power operation**.
- **Applications:** security screening, industrial inspection, hazardous object detection.

## Certifications & Achievements

- **e-Yantra Robotics Competition (eYRC) 2026-27** – Participant | IIT Bombay | Ongoing
- **NPTEL (Elite)** – Embedded System Design with ARM | IIT Kharagpur (SWAYAM)
- **e-Yantra eLSI** – Basics of Embedded Systems & Robotics, Certificate of Merit (98/100) | IIT Bombay
- **Cybersecurity Fundamentals** – Tech Mahindra Foundation | Skill India Digital Hub
- **InnovoQuest 2.0** (Innovation & Entrepreneurship Challenge) | **eLSI Sprint** (Embedded Systems & Robotics Challenge)
- **Proteus Design Suite Workshop (Cohort-I)** | GEC Kishanganj | Certificate of Participation, Aug 2026